Course Notes

Applied Abstract Algebra
Quiver Theory

Instructor:

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TA:

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Contents

Chapter 1

Lecture 1

Course Information

Main topic: Representation theory of quivers.

Office hours: Mon: 12:30-14:30.

Main book: *Ralf Schiffler, Quiver Representations*, Cambridge University Press.

References:

- *Assem, Simons, Skowronski, Elements of the Representation Theory of Associative Algebras*, Cambridge University Press.
- *Ibrahim Assem, basic representation theory of algebras.*
- *Michel Brion, Representations of quivers.*
- *Henning Krause, Representations of quivers via reflection functor.*
- *W.Crawley-Boevey, Lectures on representations of quivers.*
- *W.Crawley-Boevey, More on representations of quivers.*
- *Alexander Kirillov Jr. - Quiver Representations and Quiver Varieties (2016, American Mathematical Society).*
- Other Reference: `www.math.uni-bielefeld.de`

Grading:

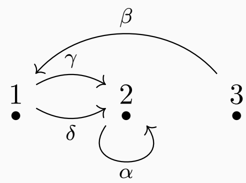
- Midterm: 30%
- Presentation: 10%
- Homework: 20%
- Final: 40%

Midterm will be on April 13th, Monday, 2026 (the 8th week).

1.1 Introduction

Definition 1.1.1. *A quiver Q is a quadruple (Q_0, Q_1, s, t), where Q_0 is a set of vertices, Q_1 is a set of arrows, and $s, t : Q_1 \to Q_0$ are two maps. For any arrow $\alpha \in Q_1$, $s(\alpha)$ is called the source of α and $t(\alpha)$ is called the target of α. For example, if $\alpha : 1 \to 2$, then $s(\alpha) = 1$ and $t(\alpha) = 2$.*

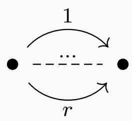
Example 1.1.2. *Let* $Q_0 = \{1, 2, 3\}$*,* $Q_1 = \{\alpha, \beta, \gamma, \delta\}$*. We have* $s(\alpha) = 2, t(\alpha) = 2$*;* $s(\beta) = 3, t(\beta) = 1$*;* $s(\gamma) = s(\delta) = 1, t(\gamma) = t(\delta) = 2$*.*



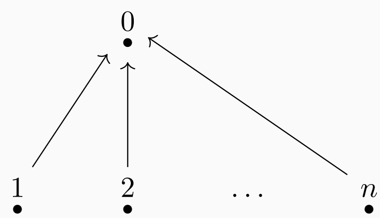
Let path $\xi = \alpha\gamma\beta$*, then we have* $s(\xi) = s(\beta) = 3$ *and* $t(\xi) = t(\alpha) = 2$*.*

Example 1.1.3. *Jordan quiver. The Jordan quiver consists of a single vertex and a single arrow starting and ending at this vertex. Formally,* $Q_0 = \{1\}$ *and* $Q_1 = \{\alpha\}$ *with* $s(\alpha) = t(\alpha) = 1$*. It is called the Jordan quiver because its representations correspond to vector spaces equipped with a linear endomorphism. The classification of such representations up to isomorphism is equivalent to the classification of square matrices up to similarity, which is given by the Jordan normal form theorem (over an algebraically closed field).*

Example 1.1.4. r*-Kronecker quiver. The* r*-Kronecker quiver has two vertices, say 1 and 2, and* r *arrows* $\alpha_1, \ldots, \alpha_r$ *from 1 to 2. Thus* $Q_0 = \{1, 2\}$*,* $Q_1 = \{\alpha_1, \ldots, \alpha_r\}$*, with* $s(\alpha_i) = 1$ *and* $t(\alpha_i) = 2$ *for all* $i = 1, \ldots, r$*. It is named after Leopold Kronecker. In the case* $r = 2$*, the representations correspond to pairs of linear maps between two vector spaces, which can be represented by matrix pencils. Kronecker solved the classification problem for these pencils.*



Example 1.1.5. n*-subspace quiver. The* n*-subspace quiver consists of* $n + 1$ *vertices* $\{0, 1, \ldots, n\}$ *and* n *arrows* $\alpha_1, \ldots, \alpha_n$ *such that* $s(\alpha_i) = i$ *and* $t(\alpha_i) = 0$ *for all* $i = 1, \ldots, n$*. It is called the* n*-subspace quiver because its representations are related to configurations of subspaces. If the linear maps associated with the arrows are injective, a representation corresponds to a collection of* n *subspaces of the vector space associated with the central vertex.*



Example 1.1.6. *Finite quivers. A quiver $Q = (Q_0, Q_1, s, t)$ is called finite if both the set of vertices Q_0 and the set of arrows Q_1 are finite sets.*

For the base field we assume $k = \bar{k}$. Usually we take $k = \mathbb{C}$.

Definition 1.1.7. *A representation M of a quiver Q consists of a vector space M_i for each vertex $i \in Q_0$ and a linear map $\phi_\alpha : M_{s(\alpha)} \to M_{t(\alpha)}$ for each arrow $\alpha \in Q_1$. We denote this representation by $M = (M_i, \phi_\alpha)_{i \in Q_0, \alpha \in Q_1}$.*

Example 1.1.8. *For example 1:*

$$\begin{tikzcd} \overset{M_1}{\bullet} \arrow[r, bend left, "\phi_\gamma"] \arrow[r, bend right, "\phi_\delta"'] & \overset{M_2}{\bullet} \arrow[loop below, "\phi_\alpha"] & \overset{M_3}{\bullet} \arrow[ll, bend right, "\phi_\beta"'] \end{tikzcd}$$

Definition 1.1.9. *A representation M is called* ***finite dimensional*** *if $\dim M_i < \infty$ for all $i \in Q_0$. The tuple $\dim M = \{\dim M_i\}_{i \in Q_0}$ is called the* ***dimension vector*** *of M. An element $m \in M$ is a tuple $\{m_i\}_{i \in Q_0}$ where $m_i \in M_i$ for each $i \in Q_0$.*

Example 1.1.10. 1-*Kronecker quiver:*

$$1 \longrightarrow 2$$

Take k to be a field. Some representations of this quiver are:

$$M: \quad k \xrightarrow{\quad 1 \quad} k \qquad M': \quad k \xrightarrow{\quad 0 \quad} k \qquad M'': \quad k \xrightarrow{\quad 0 \quad} 0$$

Another example is

$$\widetilde{M}: \quad k^2 \xrightarrow{\begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 0 \end{bmatrix}} k^3 \ .$$

For $v = (1, 0) \in k^2$, we have

$$\begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}.$$

Definition 1.1.11 (Morphism between two representations of Q)**.** *Let $M = (M_i, \phi_\alpha)_{i \in Q_0, \alpha \in Q_1}$ and $N = (N_i, \psi_\alpha)_{i \in Q_0, \alpha \in Q_1}$ be two representations of Q. A morphism $f : M \to N$ is a collection of linear maps $f_i : M_i \to N_i$ such that for every*

$\phi_\alpha : M_i \to M_j$, $\alpha \in Q_1$ *the following diagram commutes:*

$$\begin{tikzcd} M_i \arrow[r, "\phi_\alpha"] \arrow[d, "f_i"'] & M_j \arrow[d, "f_j"] \\ N_i \arrow[r, "\psi_\alpha"'] & N_j \end{tikzcd}$$

Example 1.1.12. *In example of 1-Kronecker quiver, let $f : M \to M''$, $f_1 : k \to k$ such that $f_1(1) = a$, and $f_2 : k \to 0$ is the zero map. We can check that f is a morphism. Check the following diagram:*

$$f_1 \quad \begin{tikzcd} k \arrow[r, "1"] \arrow[d, "a"'] & k \arrow[d, "0"] \\ k \arrow[r, "0"'] & 0 \end{tikzcd} \quad f_2$$

Now we establish $g : M'' \to M$. If g is a morphism, then we have $g_1 : k \to k$ such that $g_1(1) = b$, and $g_2 : 0 \to k$ is the zero map. To make the target diagram commute it forces $g_1 = 0$. Hence there are only 0 morphism form M'' to M.

$$g_1 \quad \begin{tikzcd} k \arrow[r, "0"] \arrow[d, "b"'] & 0 \arrow[d, "0"] \\ k \arrow[r, "1"'] & 0 \end{tikzcd} \quad g_2$$

$$1 \circ g_1 = 0 \Rightarrow g_1 = 0$$

Recall the definition of category:

Definition 1.1.13. *A category $\mathcal{C}$ consists of the following data:*

- *A class of objects, denoted by* $\mathrm{Ob}(\mathcal{C})$.
- *For any pair of objects $X, Y \in \mathrm{Ob}(\mathcal{C})$, a set of morphisms from X to Y, denoted by $\mathrm{Hom}_{\mathcal{C}}(X, Y)$. If $f \in \mathrm{Hom}_{\mathcal{C}}(X, Y)$, we write $f : X \to Y$.*
- *For any three objects $X, Y, Z \in \mathrm{Ob}(\mathcal{C})$, a composition map:*

$$\circ : \mathrm{Hom}_{\mathcal{C}}(Y, Z) \times \mathrm{Hom}_{\mathcal{C}}(X, Y) \to \mathrm{Hom}_{\mathcal{C}}(X, Z), \quad (g, f) \mapsto g \circ f.$$

These data must satisfy the following axioms:

1. ***Associativity**: For any morphisms $f : X \to Y$, $g : Y \to Z$, and $h : Z \to W$, we have*
$$h \circ (g \circ f) = (h \circ g) \circ f.$$
2. ***Identity**: For every object $X \in \mathrm{Ob}(\mathcal{C})$, there exists an identity morphism $\mathrm{id}_X \in$*

$\mathrm{Hom}_{\mathcal{C}}(X,X)$ *such that for any* $f : X \to Y$ *and* $g : Z \to X$,

$$f \circ \mathrm{id}_X = f \quad \textit{and} \quad \mathrm{id}_X \circ g = g.$$

Denote Rep$Q$ the category of representations of Q. Then Ob(Rep$Q$) is the class of representations of Q: $M = (M_i, \phi_\alpha)$, and $\mathrm{Hom}_{\mathrm{Rep}Q}$ is the class of morphisms $\mathrm{Hom}_{\mathrm{Rep}Q}(M,N)$ from M to N.

Let $M, N \in \mathrm{Ob}(\mathrm{Rep}Q)$. Then $\mathrm{Hom}(M,N) := \mathrm{Hom}_{\mathrm{Rep}Q}(M,N)$ is a vector space over k with operations defined component-wise: for $f, g \in \mathrm{Hom}(M,N)$ and $\lambda \in k$, let $(f+g)_i = f_i + g_i$ and $(\lambda f)_i = \lambda f_i$ for all $i \in Q_0$.

Verification: Let $M = (M_i, \phi_\alpha)$ and $N = (N_i, \psi_\alpha)$. Since f, g are morphisms, for any arrow $\alpha : i \to j$, we have $\psi_\alpha f_i = f_j \phi_\alpha$ and $\psi_\alpha g_i = g_j \phi_\alpha$.

1. **Addition:**

$$\psi_\alpha(f+g)_i = \psi_\alpha(f_i + g_i) = \psi_\alpha f_i + \psi_\alpha g_i = f_j\phi_\alpha + g_j\phi_\alpha = (f_j + g_j)\phi_\alpha = (f+g)_j\phi_\alpha.$$

   Thus $f + g \in \mathrm{Hom}(M,N)$.

2. **Scalar Multiplication:**

$$\psi_\alpha(\lambda f)_i = \psi_\alpha(\lambda f_i) = \lambda(\psi_\alpha f_i) = \lambda(f_j\phi_\alpha) = (\lambda f_j)\phi_\alpha = (\lambda f)_j\phi_\alpha.$$

   Thus $\lambda f \in \mathrm{Hom}(M,N)$.

The zero morphism $0 = (0)_{i \in Q_0}$ clearly satisfies the commutativity condition. The vector space axioms follow from the structure of linear maps between vector spaces.

Example 1.1.14. *Consider*

$$1 \longrightarrow 2$$

$\mathrm{Hom}(M, M'') = \{(a,0) | a \in k\} \simeq k$. $\mathrm{Hom}(M'', M) = \{0\}$.

Example 1.1.15. *Let* Q *be a quiver:*

$$\begin{tikzcd}
1 \arrow[dr, "\alpha"] & & \\
& 3 & 4 \arrow[l, "\gamma"'] \\
2 \arrow[ur, "\beta"] & &
\end{tikzcd}$$

Let M*:*

$$\begin{tikzcd}
k \arrow[dr, "\begin{bmatrix}1\\0\end{bmatrix}"] & & \\
& k^2 & k \arrow[l, "\begin{bmatrix}1\\1\end{bmatrix}"'] \\
k \arrow[ur, "\begin{bmatrix}0\\1\end{bmatrix}"] & &
\end{tikzcd}$$

M':

$$\begin{tikzcd}
0 \arrow[dr, "0"] & & \\
& k & 0 \arrow[l, "0"'] \\
0 \arrow[ur, "0"] & &
\end{tikzcd}$$

M'':

$$\begin{tikzcd}
k \arrow[dr, "1"] & & \\
& k & k \arrow[l, "1"'] \\
k \arrow[ur, "1"] & &
\end{tikzcd}$$

Let us check $\mathrm{Hom}(M, M')$*:*

$$\begin{tikzcd}
k \arrow[dr, "\begin{bmatrix}1\\0\end{bmatrix}"] \arrow[ddd, bend right, "0"'] & & \\
& k^2 \arrow[ddd, "\begin{bmatrix}a & b\end{bmatrix}"] & k \arrow[l, "\begin{bmatrix}1\\1\end{bmatrix}"'] \arrow[ddd, bend left, "0"] \\
k \arrow[ur, "\begin{bmatrix}0\\1\end{bmatrix}"] \arrow[ddd, bend right, "0"'] & & \\
0 \arrow[dr, "0"] & & \\
& k & 0 \arrow[l, "0"] \\
0 \arrow[ur, "0"] & &
\end{tikzcd}$$

Since $M'_1 = M'_2 = M'_4 = 0$, the maps f_1, f_2, f_4 are zero. Let $f_3 = \begin{bmatrix} a & b \end{bmatrix}$. The commutativity conditions impose:

$$\begin{cases} f_3\phi_\alpha = \psi_\alpha f_1 \implies \begin{bmatrix} a & b \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 0 \implies a = 0, \\ f_3\phi_\beta = \psi_\beta f_2 \implies \begin{bmatrix} a & b \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 0 \implies b = 0. \end{cases}$$

Thus $f_3 = 0$, *and* $\mathrm{Hom}(M, M') = 0$.
Let us check $\mathrm{Hom}(M, M'')$:

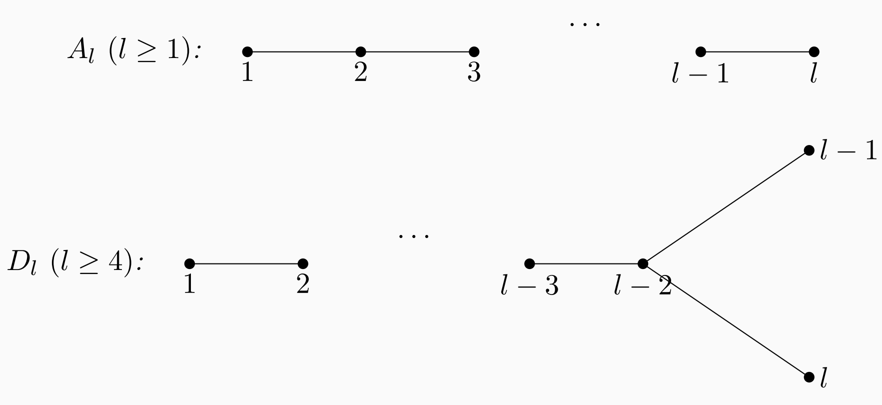
$$\begin{tikzcd}
k \arrow[dr, "\begin{bmatrix}1\\0\end{bmatrix}"] \arrow[ddd, bend right, "c"'] & & \\
& k^2 \arrow[ddd, "{[a \quad b]}"] & k \arrow[l, "\begin{bmatrix}1\\1\end{bmatrix}"'] \arrow[ddd, bend left, "e"] \\
k \arrow[ur, "\begin{bmatrix}0\\1\end{bmatrix}"] \arrow[ddd, bend right, "d"'] & & \\
0 \arrow[dr, "0"] & & \\
& k & 0 \arrow[l, "0"] \\
0 \arrow[ur, "0"] & &
\end{tikzcd}$$

Let the morphism be given by $g_1 = [c]$, $g_2 = [d]$, $g_4 = [e]$, *and* $g_3 = \begin{bmatrix} a & b \end{bmatrix}$. *The commutativity conditions are:*

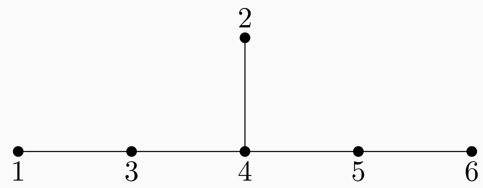
$$\begin{cases} g_3\phi_\alpha = \psi_\alpha g_1 \implies \begin{bmatrix} a & b \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = c \implies a = c, \\ g_3\phi_\beta = \psi_\beta g_2 \implies \begin{bmatrix} a & b \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = d \implies b = d, \\ g_3\phi_\gamma = \psi_\gamma g_4 \implies \begin{bmatrix} a & b \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = e \implies a + b = e. \end{cases}$$

The morphism is uniquely determined by the parameters $a, b \in k$. *Hence* $\mathrm{Hom}(M, M'') \cong k^2$.

Example 1.1.16. *Dynkin Diagrams:*



E_6:



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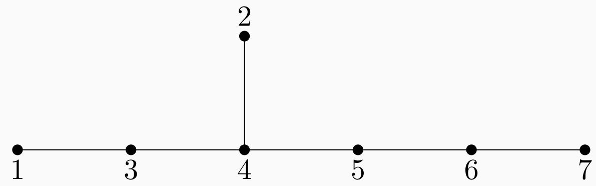
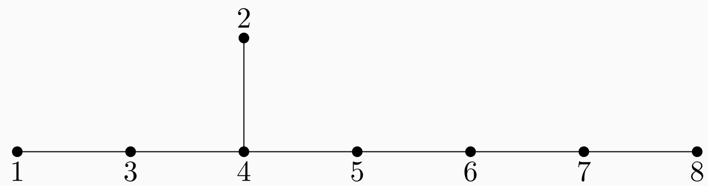


E_8:



Example 1.1.17. *2-Kronecker quiver:*

$$\begin{tikzcd} 1 \arrow[r, bend left] \arrow[r, bend right] & 2 \end{tikzcd}$$

Representations of this quiver:

$$\begin{tikzcd} V \arrow[r, bend left, "A"] \arrow[r, bend right, "B"'] & W \end{tikzcd}$$

Example 1.1.18. *Jordan quiver:*

$$\begin{tikzcd} 1 \arrow[loop above] \end{tikzcd}$$

Representations of this quiver:

$$\begin{tikzcd} V \arrow[loop above, "A"] \end{tikzcd}$$

$A : V \to V$, the matrix representation of A is similar to a Jordan normal form (when $k = \bar{k}$).

Chapter 2

Lecture 2

Recall the definition:

Definition 2.0.1. *Let $Q = (Q_0, Q_1)$ be a quiver. A representation of Q is a collection*

$$M = (V_i, \varphi_\alpha)_{i\in Q_0, \alpha\in Q_1},$$

where each V_i is a vector space and each arrow $\alpha : i \to j$ is assigned a linear map

$$\varphi_\alpha \in \mathrm{Hom}(V_i, V_j).$$

Example 2.0.2. *Consider the 2-Kronecker quiver*

$$\begin{tikzcd} 1 \arrow[r, bend left] \arrow[r, bend right] & 2 \end{tikzcd}$$

and the representations

$$M : \qquad k \overset{\begin{bmatrix}1\\0\end{bmatrix}}{\underset{\begin{bmatrix}0\\1\end{bmatrix}}{\rightrightarrows}} k^2$$

and

$$N : \qquad k^2 \overset{\begin{bmatrix}1&1\\0&1\end{bmatrix}}{\underset{\begin{bmatrix}1&1\\1&1\end{bmatrix}}{\rightrightarrows}} k^2$$

Denote $M = (V_i, \varphi_\alpha)_{i\in Q_0,\alpha\in Q_1}$ and $N = (U_i, \psi_\alpha)_{i\in Q_0,\alpha\in Q_1}$. Consider the morphism $f : M \to N$. $f = (f_i)_{i\in Q_0}$. $f_i : V_i \to U_i$ and for each $\alpha : i \to j$ in Q, the following *diagram commutes:*

$$\begin{tikzcd} V_i \arrow[r, "\varphi_\alpha"] \arrow[d, "f_i"'] & V_j \arrow[d, "f_j"] \\ U_i \arrow[r, "\psi_\alpha"'] & U_j \end{tikzcd}$$

We use $\mathrm{Rep}(Q)$ *to denote the category of representations of* Q*. For our example, we have*

$$f : M \to N : f = (f_1, f_2) :$$

Let $f = (f_1, f_2) : M \to N$ *be a morphism. Write*

$$f_1 = \begin{bmatrix} a \\ b \end{bmatrix}, \qquad f_2 = \begin{bmatrix} c & d \\ e & f \end{bmatrix}.$$

Then the following diagram commutes:

$$\begin{tikzcd} k \arrow[r, shift left, "\begin{bmatrix} 1 \\ 0 \end{bmatrix}"] \arrow[r, shift right, "\begin{bmatrix} 0 \\ 1 \end{bmatrix}"'] \arrow[d, "f_1 = \begin{bmatrix} a \\ b \end{bmatrix}"'] & k^2 \arrow[d, "f_2 = \begin{bmatrix} c & d \\ e & f \end{bmatrix}"] \\ k^2 \arrow[r, shift left, "\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}"] \arrow[r, shift right, "\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}"'] & k^2 \end{tikzcd}$$

Commutativity along the two arrows gives

$$f_2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} c \\ e \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} f_1 = \begin{bmatrix} a+b \\ b \end{bmatrix},$$

and

$$f_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} d \\ f \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} f_1 = \begin{bmatrix} a+b \\ a+b \end{bmatrix}.$$

Hence

$$c = d = f = a + b, \qquad e = b,$$

so

$$f = \left(\begin{bmatrix} a \\ b \end{bmatrix}, \begin{bmatrix} a+b & a+b \\ b & a+b \end{bmatrix} \right), \qquad \dim \mathrm{Hom}(M, N) = 2.$$

This illustrates how morphisms in $\mathrm{Rep}(Q)$ *are determined by commutative diagrams.*

Definition 2.0.3. *If $M = (V_i, \varphi_\alpha)$ and $N = (U_i, \psi_\alpha)$ are representations of Q, then their direct sum is*

$$M \oplus N = \left(V_i \oplus U_i, \begin{bmatrix} \varphi_\alpha & 0 \\ 0 & \psi_\alpha \end{bmatrix} \right).$$

Inductively, one may form finite direct sums $M_1 \oplus \cdots \oplus M_r$.

Definition 2.0.4. *A representation M is called indecomposable if it cannot be written as a direct sum of two non-zero representations of Q.*

Remark 2.0.5. *For quiver representations, indecomposable and irreducible are different notions; they do not have to coincide.*

Example 2.0.6. *Consider the quiver*

$$1 \longrightarrow 2 \longleftarrow 3$$

and the following representations:

$$M : \quad k \xrightarrow{1} k \xleftarrow{0} 0$$

$$N : \quad k^2 \xrightarrow{\begin{bmatrix} 0 \\ 1 \end{bmatrix}} k^2 \xleftarrow{\begin{bmatrix} 1 \\ 1 \end{bmatrix}} k$$

Then

$$M \oplus N : \quad k \oplus k^2 \xrightarrow{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}} k \oplus k^2 \xleftarrow{\begin{bmatrix} 0 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix}} 0 \oplus k$$

Indeed, identifying

$$k \oplus k^2 \cong k^3, \qquad k \oplus k^2 \cong k^3, \qquad 0 \oplus k \cong k,$$

we may rewrite this as

$$k^3 \xrightarrow{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}} k^3 \xleftarrow{\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}} k$$

Example 2.0.7. *Consider the Jordan quiver:*

$$\begin{tikzcd} \bullet \arrow[loop above] \end{tikzcd}$$

Let A, B, C be three $n \times n$ matrices. Where C denotes an isomorphism (C is invertible). We have the following diagram:

$$
\begin{tikzcd}
\overset{\bullet}{V} \arrow[loop above, "A"] \arrow[d, "C"'] \\
\overset{\bullet}{V} \arrow[loop below, "B"]
\end{tikzcd}
\qquad
\begin{tikzcd}
V \arrow[r, "A"] \arrow[d, "C"'] & V \arrow[d, "C"] \\
V \arrow[r, "B"'] & V
\end{tikzcd}
$$

Which shows that A and B are similar: $A = C^{-1}BC$. Hence the isomorphism classes are controlled by Jordan normal form, and the indecomposable representations correspond exactly to single Jordan blocks.

To describe $\operatorname{Rep} Q$ we will need to describe all isomorphism classes of indecomposable representations.

Theorem 2.0.8 (Krull–Schmidt)**.** *Any finite representation $M \in \operatorname{Rep} Q$ can be decomposed into finite number of indecomposable representations and its decomposition is unique up to order.*

2.1 Kernel and Cokernel

Recall in linear algebra, we have the following definitions:

Definition 2.1.1. *Let $f : V \to W$ be a linear map. The kernel of f is the set of all vectors $v \in V$ such that $f(v) = 0$.*

Definition 2.1.2. *Let $f : V \to W$ be a linear map. The image of f is the set of all vectors $w \in W$ such that $f(v) = w$ for some $v \in V$.*

Definition 2.1.3. *Let $f : V \to W$ be a linear map. The cokernel of f is the quotient space $W/\operatorname{Im} f$.*

Let $f : M \to N$ ba a morphism from M to N where $M, N \in \operatorname{Rep} Q$. $M = (V_i, \varphi_\alpha)$ and $N = (U_i, \psi_\alpha)$. Define $\ker f \in \operatorname{Rep} Q$ as (L_i, χ_α) where $L_i = \ker f_i$ and $\chi_\alpha = \varphi_\alpha|_{L_i}$. $\forall v_i \in L_i$, $\chi_\alpha(v_i) = \varphi_\alpha(v_i)$.

We want to show that $\ker f$ is a representation in $\operatorname{Rep} Q$. i.e. $X_\alpha : L_i \to L_j$ for $\alpha : i \to j$

Chapter 3

Lecture 3

Example 3.0.1. $1 \longrightarrow 2$ *We have trivial representation*

$$0 \xrightarrow{0} 0$$

The following 3 representations are non-trivial

$$k \xrightarrow{0} 0 \quad 0 \xrightarrow{0} k \quad k \xrightarrow{1} k$$

Denoted by S_1, S_2, M.
First two representations are irriducible and indecomposable, the last one is decomposable. Recall that for a morophism $f : M \to N$*, we have induced representation*

$$\ker f \xhookrightarrow{incl.} M$$

And a subrepresentation of M *is representation* L *such that* $f : L \to M$ *is a monomorphism.*
Now consider

$$\begin{tikzcd} k \arrow[r, "0"] \arrow[d, "a"'] & 0 \arrow[d, "b"] \\ k \arrow[r, "1"] & k \end{tikzcd}$$

Then a=b=0, it is impossible to construct inclusion.
But if we consider

$$\begin{tikzcd} 0 \arrow[r, "0"] \arrow[d, "a"'] & k \arrow[d, "b"] \\ k \arrow[r, "1"] & k \end{tikzcd}$$

Then $a = b \cdot 0 = 0$ *and* b *is arbitrary (then we have a monomorphism). Hence we come to the conclusion that* $\mathrm{Hom}(S_1, M) = 0$ *and* $\mathrm{Hom}(S_2, M) = k$*. In other words* S_2 *is a subrepresentation of* M *and hence* M *is reducible.*
But if S_2 *is a direct summand of* M*, then consider*

Definition 3.0.2 (equivalent definition of $\ker f$)**.** *Let $f : M \to N$ be a morphism of representations. We call* $\ker f$ *a representation $g : L \to M$ such that $fg = 0$ such that for any honomorphism $h : X \to M$ with $fh = 0$, there exists a unique morphism $u : X \to L$ such that $gu = h$. In other words, the following diagram commutes:*

$$\begin{tikzcd}
X \arrow[r, dashed, "u"] \arrow[dr, "h"'] & L \arrow[d, "g"] & \\
 & M \arrow[r, "f"] & N
\end{tikzcd}$$

Let us show that $\ker f = (\ker f_i = L_i, \varphi_\alpha|_{L_i})$ is equivalent to the diagram representation $L = \ker f, g_i = ind.$ i.e. $\ker f_i \xhookrightarrow{incl.} M_i$ where $f_i g_i = 0$.

Let $h : X \to M$ be a morphism such that $fh = 0$. Let $u : X \to L$. Consider $(M_i, \varphi_\alpha), (L_i, \psi_\alpha)$ and (X_i, ν_α). $\forall x_i \in X_i$, $f_i h_i(x_i) = 0$ i.e. $h_i(x_i) \in \ker f_i = L_i$. Define $u_i(x_i) = h_i(x_i)$.

We have to check that u is a morphism and $h = gu$. Consider the following diagram:

$$\begin{tikzcd}
X_i \arrow[r, "\nu_\alpha"] \arrow[d, "u_i"'] & X_j \arrow[d, "u_j"] \\
L_i \arrow[r, "\psi_\alpha"] & L_j
\end{tikzcd}$$

Analogously we have equivalent definition of $\operatorname{coker} f$.

Definition 3.0.3. *Let $f : M \to N$ be a morphism of representations. $f : f_i \in$* $\mathrm{Hom}(M_i, N_i)_{i \in Q_0}$. $\operatorname{coker} f = (N_i/f_i(M_i), \chi_\alpha(n_i + f_i(M_i)) = \psi_\alpha(n_i), f_j(M_j))$. *Equivalently,* $\operatorname{coker} f$ *is a representation $g : N \to H$ such that $gf = 0$ and for any morphism $h : N \to Y$ with $hf = 0$, there exists a unique morphism $u : H \to Y$ such that $ug = h$. In other words, the following diagram commutes:*

Exercise 3.0.4. *Prove the quivalence of the two definitions of* $\operatorname{coker} f$*.*

Theorem 3.0.5 (firsi isomorphism theorem)**.** *If $f : M \to N$ is a morphism of representations, then* $\operatorname{im} f \cong M/\ker f$*.*

Proof. $(\operatorname{im} f)_i = \operatorname{im} f_i \subseteq N_i$ and $\operatorname{im} f = (\operatorname{im} f_i, \psi_\alpha|_{\operatorname{im} f_i})$. Check that this is a representation. Let $(M/\ker f)_i = M_i/\ker f_i$, $M/\ker f = (M_i/\ker f_i, \overline{\varphi_\alpha})$ where $\overline{\varphi_\alpha}$ is defined as $\overline{\varphi_\alpha}(m_i + \ker) = \varphi_\alpha(m_i) + \ker f_j$.

From the isomorphism theorem for vector spaces, $\bar{f}_i(m_i + \ker f_i) := f_i(m_i)$, $\bar{f}_i : M_i/\ker f_i \to \operatorname{im} f_i$. We need to check that $\bar{f} : M/\ker f \cong \operatorname{im} f$.

Consider the following diagram:

$$\begin{tikzcd}
M_i/\ker f_i \arrow[r, "\overline{\varphi_\alpha}"] \arrow[d, "u_i"'] & M_j/\ker f_j \arrow[d, "u_j"] \\
\operatorname{im} f_i \arrow[r, "\psi_\alpha|_{\operatorname{im} f_i}"] & \operatorname{im} f_j
\end{tikzcd}$$

Check the commutativity of the diagram: □

These all together shows that $\operatorname{Rep} Q$ is abelian category. More precisely,

Definition 3.0.6. *We call a category $\mathcal{C}$ abelian k-category if the following conditions are satisfied:*

1. *$\mathcal{C}$ is k-categor, for any $X, Y \in \mathrm{Ob}(\mathcal{C})$, $\mathrm{Hom}_{\mathcal{C}}(X, Y)$ is a vector space over k.*
2. *$\mathcal{C}$ is a additive category, $\mathcal{C}$ has direct sums and zero object.*
3. *Each morphism $f : X \to Y$ has kernel K and cokernel H. $f : M \to N$, $i : K \to M$, $p : N \to H$. and $\operatorname{coker} i \cong \ker p$.*

Definition 3.0.7. *Let $M \longrightarrow N \longrightarrow P$ be a sequence of morphisms in $\operatorname{Rep} Q$. We say that this sequence is exact if $\ker p = \operatorname{im} f$.*

Definition 3.0.8. *We call a sequence of morphisms in $\operatorname{Rep} Q$ a short exact sequence if it is exact and of the form*

$$0 \longrightarrow L \xrightarrow{f} M \xrightarrow{g} N \longrightarrow 0$$

Where f is injective, g is surjective $\operatorname{im} f = \ker g$

For $f : M \to N$ we have exact sequence:

$$0 \longrightarrow \ker f \xrightarrow{i} M \xrightarrow{f} N \xrightarrow{p} \operatorname{coker} f \longrightarrow 0$$

Where i is an inclusion and p is a projection.

Or we have short exact sequence:

$$0 \longrightarrow \ker f \xrightarrow{i} M \xrightarrow{p} M/\ker f \longrightarrow 0$$

Example 3.0.9. *picture*

Definition 3.0.10. *A morphism $f : L \to M$ a section if there exists a morphism $h : M \to L$ such that $hf = 1_L$. A morphism $g : M \to N$ is a retraction if there exists a morphism $h : N \to M$ such that $gh = 1_N$.*

Definition 3.0.11. *We say that a short exact sequence*

$$0 \longrightarrow L \xrightarrow{f} M \xrightarrow{g} N \longrightarrow 0$$

splits if f has a retraction or equivalently f is a section.

Theorem 3.0.12. *Let*

$$0 \longrightarrow L \xrightarrow{f} M \xrightarrow{g} N \longrightarrow 0$$

be a short exact sequence in $\operatorname{Rep} Q$. *Then*

1. f *is a section if and only if* g *is a retraction.*

2. *If* f *is a seciton then* $\ker g = \operatorname{im} f$ *is a direct summand of* M.

Proof. $(\Rightarrow)$: Suppose f is a section. $\exists h : M \to L$ such that $hf = 1_L$. We will define $h' : N \to M$ and show that $gh' = 1_N$. $\forall n \in N$, since g is surjective $\exists m$ such that $n = g(m)$. $h'(n) = m - f(h(m))$ where $h' : N \to M$. We have to show that h' is well-defined.

Suppose m, m' such that $n = g(m) = g(m')$. We have to show that $m' - f(h(m')) = m - f(h(m))$. $m' - m = f(h(m')) - f(h(m)) \in \operatorname{im} f$, $\exists l \in L$ such that $f(l) = m' - m =$. Now:

$$m' - f(h(m')) - m + f(h(m)) = m' - m - f(h(m' - m)) = m' - m - f(h(f(l))) = m' - m - f(l) = 0$$

$\square$